

Hardware, Binary System & Data Units – Page 490

Source: Lucent's General Knowledge – Computer

Computer: Hardware, Binary System, Data Measurement & Logic Gates

50 Questions • 30 Minutes

Instructions:

- Each question has 4 options (A, B, C, D).
- The correct answer is shown on the right side of each question.
- Detailed explanations with textbook page references are at the end.

Questions

Q1. A computer has two main parts:

- A. Monitor and Keyboard
- B. Hardware and Software
- C. CPU and Memory
- D. Input and Output

Ans: B

Q2. Hardware comprises of the physical units of a computer system. These units constitute the:

- A. Software
- B. Hardware of a computer
- C. Programs
- D. Data

Ans: B

Q3. A computer has three main units:

- A. ALU, CU, Memory
- B. Input unit, Processing unit, Output unit
- C. Keyboard, Mouse, Monitor
- D. CPU, RAM, HDD

Ans: B

Q4. The computer has its own internal:

- A. Voice
- B. Language
- C. Music
- D. Video

Ans: B

Q5. The computer is essentially made of electronic components that can generate either a low (0 volt) or a high (say 5 volts). These states are represented by:

- A. Alphabets
- B. Binary digits (bits) - 1s and 0s
- C. Decimal numbers
- D. Roman numerals

Ans: B

Q6. Computers always work with:

- A. Decimal numbers
- B. Bits
- C. Words
- D. Pictures

Ans: B

Q7. Computer scientists use combination of how many bits to represent various symbols?

- A. 4
- B. 8
- C. 16
- D. 32

Ans: B

Q8. Using 8 bits, the computer can represent how many different symbols? A. 128 B. 256 C. 64 D. 512	Ans: B
Q9. A combination of 8 bits is called a: A. Nibble B. Byte C. Word D. Kilobyte	Ans: B
Q10. In computers, information is represented using multiples of 8 (2^3) bits. Higher units are expressible in multiples of: A. 2^2 B. 2^3 C. 10 D. 100	Ans: B
Q11. The power of two closest to a kilo (1000) is $2^{10} =$: A. 512 B. 1024 C. 2048 D. 256	Ans: B
Q12. 1 Mega Byte is equal to: A. 1024 Bytes B. 1 million bytes or 1000 Kilo Bytes C. 1 billion bytes D. 512 KB	Ans: B
Q13. A Giga Byte refers to approximately: A. 1000 bytes B. 1000 megabytes or 1 million kilobytes C. 100 megabytes D. 10 megabytes	Ans: B
Q14. 4 Bit = : A. 1 Byte B. 1 Nibble C. 1 KB D. 1 Word	Ans: B

Q15. 2 Nibble (8 Bit) = : A. 1 Nibble B. 1 Byte C. 1 KB D. 1 MB	Ans: B
Q16. 1024 Byte = : A. 1 MB B. 1 Kilo Byte (KB) C. 1 GB D. 1 Nibble	Ans: B
Q17. 1024 KB = : A. 1 GB B. 1 Mega Byte (MB) C. 1 TB D. 1 Byte	Ans: B
Q18. 1024 MB = : A. 1 TB B. 1 Giga Byte (GB) C. 1 PB D. 1 KB	Ans: B
Q19. 1024 GB = : A. 1 PB B. 1 Tera Byte (TB) C. 1 MB D. 1 KB	Ans: B
Q20. 1024 TB = : A. 1 GB B. 1 Peta Byte (PB) C. 1 MB D. 1 EB	Ans: B
Q21. The basic elements of computers that can signal a 1 or 0 are called: A. Transistors B. Flip-flops C. Resistors D. Capacitors	Ans: B
Q22. Modern computers use a very tiny set of flip-flops known as: A. RAM B. Register C. Cache D. ROM	Ans: B

Q23. The entire independent circuits can be designed on a small piece of semi conductor material like: A. Copper B. Silicon C. Gold D. Iron	Ans: B
Q24. Silicon is obtained from sand and is a poor conductor of electricity. Miniature circuits made on silicon are called: A. Transistors B. Integrated Circuits (IC) C. Resistors D. Capacitors	Ans: B
Q25. By 1971, engineers were able to build a complete computer on a single chip of silicon called: A. RAM chip B. Microprocessor C. ROM chip D. GPU	Ans: B
Q26. The computer converts all decimal numbers into: A. Octal numbers B. Binary numbers or combinations of bits C. Hexadecimal numbers D. Roman numerals	Ans: B
Q27. The internal circuits that perform mathematical operations on bits are usually made of: A. Resistors B. Logic gates C. Capacitors D. Inductors	Ans: B
Q28. The three basic logic gates are: A. ADD, SUB, MUL B. AND, OR and NOT C. IF, THEN, ELSE D. NAND, NOR, XOR	Ans: B
Q29. A computer is organized into three basic units: A. Hardware, Software, Data B. CPU, Memory Unit, Input/Output Unit C. Monitor, Keyboard, Mouse D. RAM, ROM, Cache	Ans: B

Q30. The CPU is the part of a computer that performs the main function of: A. Data storage B. Information processing C. Printing D. Networking	Ans: B
Q31. While earliest machines were 8-bit, contemporary computers can work with: A. 4 bits B. 16, 32 and even 64 bits C. Only 8 bits D. 128 bits	Ans: B
Q32. The earliest computers were designed so that their components could work with how many bits at a time? A. 4 B. 8 C. 16 D. 32	Ans: B
Q33. Hertz is a unit of: A. Power B. Frequency C. Length D. Weight	Ans: B
Q34. 1 MHz equals: A. 1000 ticks per second B. 1 million ticks per second C. 1 billion ticks per second D. 100 ticks per second	Ans: B
Q35. 1 GHz = : A. 100 MHz B. 1000 MHz C. 10 MHz D. 10000 MHz	Ans: B
Q36. The ticks per second of the internal clock are measured in: A. Watts B. Megahertz and Gigahertz C. Meters D. Kilograms	Ans: B

Q37. Higher the clock-speed of CPU: A. Slower the computer B. Faster the computer C. No effect D. Less memory	Ans: B
Q38. A flip-flop is a simple: A. Mechanical device B. Electrical device C. Optical device D. Chemical device	Ans: B
Q39. The most important characteristic of registers is that binary digits can be stored in them using: A. Magnets B. Certain voltages C. Light signals D. Sound waves	Ans: B
Q40. The binary number system is similar to decimal system. The only difference is in: A. Operations B. The number of digits used C. Speed D. Size	Ans: B
Q41. In binary system, only which digits are used? A. 0 to 9 B. 0 and 1 C. 0 to 7 D. 0 to F	Ans: B
Q42. 8 bits are enough to cover our entire range of alphabets, numbers and special characters like \$, @, +, ■. This is because $2^8 =$: A. 128 B. 256 C. 512 D. 64	Ans: B
Q43. To build complex information like paragraphs, we need larger numbers of bytes called: A. Bits and Nibbles B. Kilobyte, Megabyte and Gigabyte C. Only Nibbles D. Only Bits	Ans: B

Q44. Logic gates are components that generate a 1 or a 0 depending on: A. Time B. The input C. Temperature D. Pressure	Ans: B
Q45. Computer scientists use a special convention: A high is represented by '1' and a low is represented by: A. 2 B. 0 C. 9 D. 5	Ans: B
Q46. It is difficult to talk always in terms of currents and voltages. Therefore computer scientists represent information using: A. Analog signals B. Binary convention (1s and 0s) C. Sound waves D. Light pulses	Ans: B
Q47. The term 'binary' refers to something that has: A. Three parts B. Two parts C. Four parts D. One part	Ans: B
Q48. Linus released Linux under: A. MIT License B. General Public Licence (open and free to all) C. Proprietary license D. Commercial license	Ans: B
Q49. A computer may be used to control purely mechanical action. It has: A. Only software B. Two main parts: Hardware and Software C. Only hardware D. Only data	Ans: B
Q50. The computer supplies processed information back to the users using special: A. Input devices B. Output devices C. Storage devices D. Network devices	Ans: B

Answer Key & Explanations

Q1. Answer: B — Hardware and Software

Page 490: It has two main parts: (i) Hardware and (ii) Software.

Topic: Hardware

Q2. Answer: B — Hardware of a computer

Page 490: A computer has three main units: Input unit, Processing unit, Output unit. These constitute the hardware.

Topic: Hardware

Q3. Answer: B — Input unit, Processing unit, Output unit

Page 490: A computer has three main units: 1. Input unit, 2. Processing unit and 3. Output unit.

Topic: Hardware

Q4. Answer: B — Language

Page 490: The computer has its own internal 'language'.

Topic: Hardware

Q5. Answer: B — Binary digits (bits) - 1s and 0s

Page 490: A high is represented by '1' and a low by '0'. The 1s and 0s are known as binary digits or 'bits'.

Topic: Binary System

Q6. Answer: B — Bits

Page 490: Computers always work with bits. They do not understand any other form.

Topic: Binary System

Q7. Answer: B — 8

Page 490: Computer scientists use combination of 8 bits taken together to represent various symbols.

Topic: Binary System

Q8. Answer: B — 256

Page 490: Because every bit can take one of 2 possible values, using 8 bits, the computer can represent 256 different symbols.

Topic: Binary System

Q9. Answer: B — Byte

Page 490: Such a combination of 8 bits is called a byte.

Topic: Binary System

Q10. Answer: B — 2^3

Page 490: Since 8 bits are the smallest unit of information, higher units are expressible in multiples of 2^3 .

Topic: Data Units

Q11. Answer: B — 1024

Page 490: The power of two closest to a kilo (1000) is $2^{10} = 1024$.

Topic: Data Units

Q12. Answer: B — 1 million bytes or 1000 Kilo Bytes

Page 490: Mega Bytes is used in conventional sense equal to 1 million bytes or 1000 Kilo Bytes.

Topic: Data Units

Q13. Answer: B — 1000 megabytes or 1 million kilobytes

Page 490: A Giga Byte refers to 1000 megabytes approximately or 1 million kilobytes.

Topic: Data Units

Q14. Answer: B — 1 Nibble

Page 490: 4 Bit = 1 Nibble.

Topic: Data Units

Q15. Answer: B — 1 Byte

Page 490: 2 Nibble (8 Bit) = 1 Byte.

Topic: Data Units

Q16. Answer: B — 1 Kilo Byte (KB)

Page 490: 1024 Byte = 1 Kilo Byte (KB).

Topic: Data Units

Q17. Answer: B — 1 Mega Byte (MB)

Page 490: 1024 KB = 1 Mega Byte (MB).

Topic: Data Units

Q18. Answer: B — 1 Giga Byte (GB)

Page 490: 1024 MB = 1 Giga Byte (GB).

Topic: Data Units

Q19. Answer: B — 1 Tera Byte (TB)

Page 490: 1024 GB = 1 Tera Byte (TB).

Topic: Data Units

Q20. Answer: B — 1 Peta Byte (PB)

Page 490: 1024 TB = 1 Peta Byte (PB).

Topic: Data Units

Q21. Answer: B — Flip-flops

Page 490: The basic elements that can signal a 1 or 0 are called flip-flops. It is a simple electrical device.

Topic: Hardware

Q22. Answer: B — Register

Page 490: Modern computers use a very tiny set of flip-flops known as a register.

Topic: Hardware

Q23. Answer: B — Silicon

Page 490: The entire independent circuits can be designed on a small piece of semi conductor material like silicon.

Topic: Hardware

Q24. Answer: B — Integrated Circuits (IC)

Page 490: Silicon from sand is a poor conductor. Such miniature circuits are called Integrated Circuits (IC).

Topic: Hardware

Q25. Answer: B — Microprocessor

Page 490: By 1971, engineers built a complete computer on a single silicon chip called the microprocessor.

Topic: Hardware

Q26. Answer: B — Binary numbers or combinations of bits

Page 490: The computer converts all decimal numbers into binary numbers or combinations of bits.

Topic: Binary System

Q27. Answer: B — Logic gates

Page 490: The internal circuits are usually made of two or more logic gates. Logic gates generate a 1 or 0 depending on input.

Topic: Logic Gates

Q28. Answer: B — AND, OR and NOT

Page 490: The three basic logic gates are AND, OR and NOT.

Topic: Logic Gates

Q29. Answer: B — CPU, Memory Unit, Input/Output Unit

Page 490: A computer is organized into three basic units: 1. CPU, 2. Memory Unit (MU), 3. Input/Output Unit.

Topic: Hardware

Q30. Answer: B — Information processing

Page 490: The CPU is the part that performs the main function of information processing.

Topic: CPU

Q31. Answer: B — 16, 32 and even 64 bits

Page 490: While earliest machines were 8-bit, contemporary computers can work with 16, 32 and even 64 bits. This is called data width.

Topic: Hardware

Q32. Answer: B — 8

Page 490: The earliest computers were designed so that their components could work with 8 bits at a time.

Topic: Hardware

Q33. Answer: B — Frequency

Page 490: Hertz is a unit of frequency.

Topic: Hardware

Q34. Answer: B — 1 million ticks per second

Page 490: 1 MHz = 1 million 'ticks' per second.

Topic: Hardware

Q35. Answer: B — 1000 MHz

Page 490: 1 GHz = 1000 MHz.

Topic: Hardware

Q36. Answer: B — Megahertz and Gigahertz

Page 490: The ticks per second of the internal clock are measured in megahertz and gigahertz.

Topic: Hardware

Q37. Answer: B — Faster the computer

Page 490: Higher the clock-speed, faster the computer.

Topic: Hardware

Q38. Answer: B — Electrical device

Page 490: Flip-flop is a simple electrical device and can either be a '0' or a '1', meaning current flow is either inward or outward.

Topic: Hardware

Q39. Answer: B — Certain voltages

Page 490: The most important characteristic of registers is that binary digits can be stored in them using certain voltages.

Topic: Hardware

Q40. Answer: B — The number of digits used

Page 490: The binary number system is similar to decimal. Only difference is in the number of digits used (2 vs 10).

Topic: Binary System

Q41. Answer: B — 0 and 1

Page 490: Binary uses only 0 and 1, while decimal uses 0-9.

Topic: Binary System

Q42. Answer: B — 256

Page 490: 8 bits = 256 combinations, enough for all alphabets, numbers and special characters.

Topic: Binary System

Q43. Answer: B — Kilobyte, Megabyte and Gigabyte

Page 490: For complex information we need larger number of bytes - kilobyte, megabyte and gigabyte.

Topic: Data Units

Q44. Answer: B — The input

Page 490: Logic gates are components that generate a 1 or a 0 depending on the input.

Topic: Logic Gates

Q45. Answer: B — 0

Page 490: A high is symbolically represented by a '1' and a low is represented by a '0'.

Topic: Binary System

Q46. Answer: B — Binary convention (1s and 0s)

Page 490: Computer scientists use binary convention: high=1, low=0.

Topic: Binary System

Q47. Answer: B — Two parts

Page 490: The term 'binary' refers to something that has two parts.

Topic: Binary System

Q48. Answer: B — General Public Licence (open and free to all)

Page 490: Linus released Linux under General Public Licence, making it open and free to all.

Topic: Software

Q49. Answer: B — Two main parts: Hardware and Software

Page 490: A computer may be used to control purely mechanical action. It has two main parts: Hardware and Software.

Topic: Hardware

Q50. Answer: B — Output devices

Page 490: The computer supplies processed information back to the users using special output devices.

Topic: Hardware